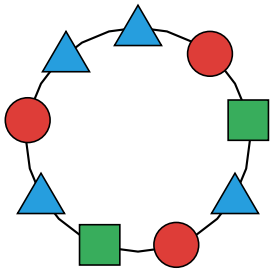
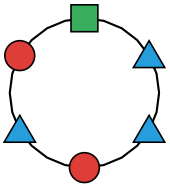
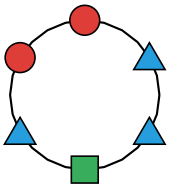
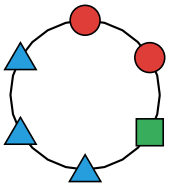
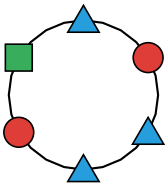
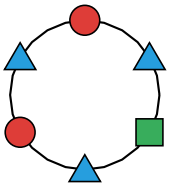
Preview: set-01/by_level/Grade_03-04_Ecolier/y2025_q21 (v1)

Closed loops: groups of neighbours can wrap around the start; recognise the same bracelet turned

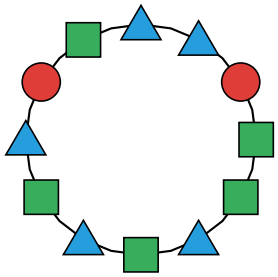
Part A: Each question is worth 3 points

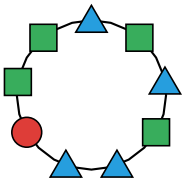
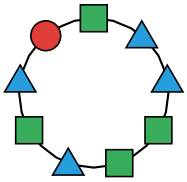
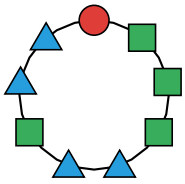
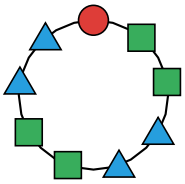
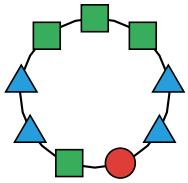
1. The picture shows a bracelet with circles (red), squares (green) and triangles (blue). Leo removes three neighbouring stones, one of each shape. Which bracelet can Leo get?



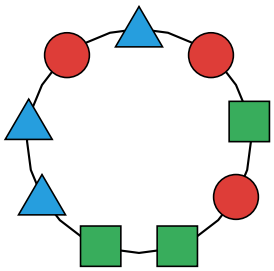
A.  ***B.***  ***C.***  ***D.***  ***E.*** 

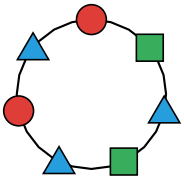
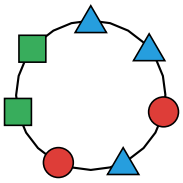
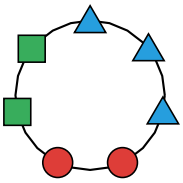
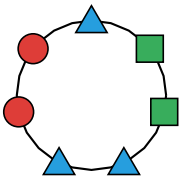
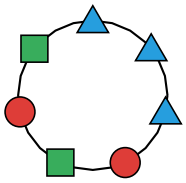
2. The picture shows a bracelet with circles (red), squares (green) and triangles (blue). Tom removes three neighbouring stones, one of each shape. Which bracelet can Tom get?



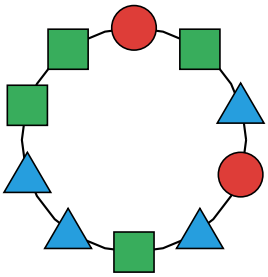
A.  ***B.***  ***C.***  ***D.***  ***E.*** 

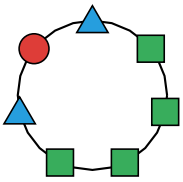
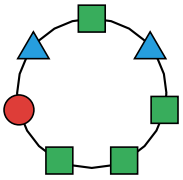
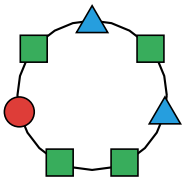
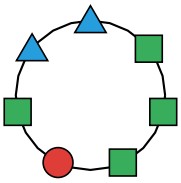
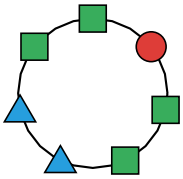
3. The picture shows a bracelet with circles (red), squares (green) and triangles (blue). Tom removes two neighbouring stones: one square and one circle. Which bracelet can Tom get?



A.  ***B.***  ***C.***  ***D.***  ***E.*** 

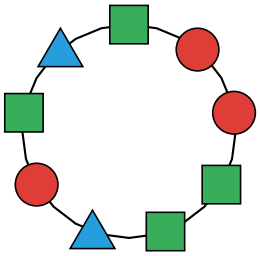
4. The picture shows a bracelet with circles (red), squares (green) and triangles (blue). Ben removes three neighbouring stones: two triangles and one circle. Which bracelet can Ben get?

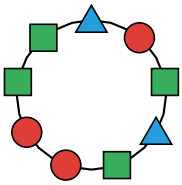
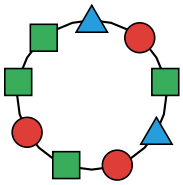
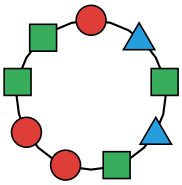
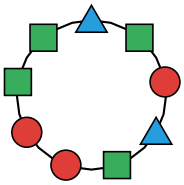
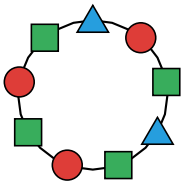


A.  ***B.***  ***C.***  ***D.***  ***E.*** 

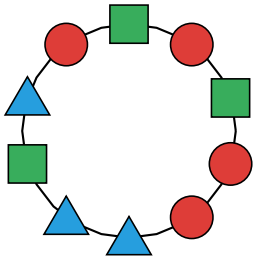
Part B: Each question is worth 4 points

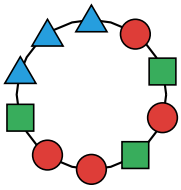
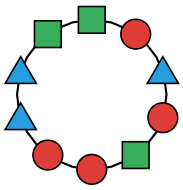
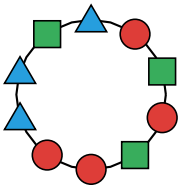
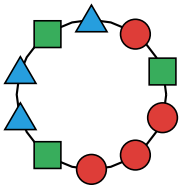
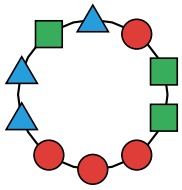
5. Anna has this bracelet with circles (red), squares (green) and triangles (blue) and turns it around a bit. Which picture can show the bracelet now?



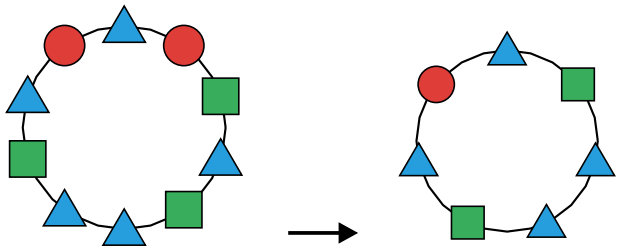
A.  ***B.***  ***C.***  ***D.***  ***E.*** 

6. Leo has this bracelet with circles (red), squares (green) and triangles (blue) and turns it around a bit. Which picture can show the bracelet now?



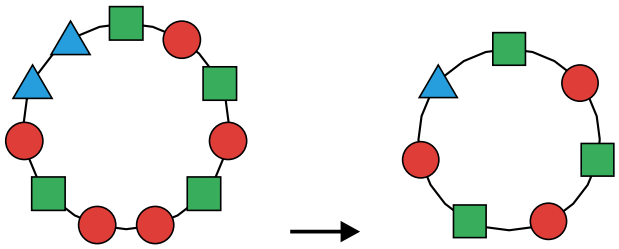
A.  ***B.***  ***C.***  ***D.***  ***E.*** 

7. Anna takes away 3 neighbouring stones from the bracelet on the left. Then the bracelet looks like the one on the right. Which stones did Anna take away?



A. ●▲▲ ***B.*** ●●▲ ***C.*** ●■▲ ***D.*** ■▲▲ ***E.*** ●●■

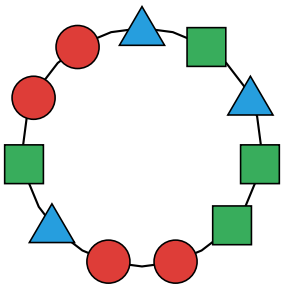
8. Ben takes away 4 neighbouring stones from the bracelet on the left. Then the bracelet looks like the one on the right. Which stones did Ben take away?



A. ●●●■ ***B.*** ●●■■ ***C.*** ●■▲▲ ***D.*** ●●■▲ ***E.*** ●●●▲

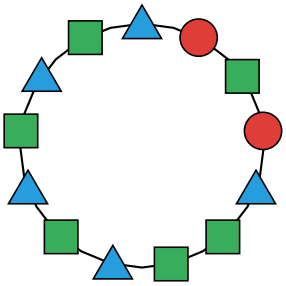
Part C: Each question is worth 5 points

9. Ben has this bracelet with circles (red), squares (green) and triangles (blue) and wants to take away three **neighbouring** stones, one of each shape. In how many different places of the bracelet is this possible?



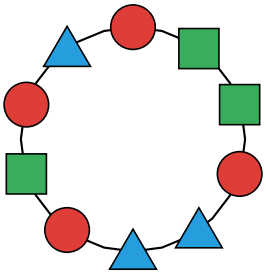
A. 1 ***B.*** 2 ***C.*** 3 ***D.*** 4 ***E.*** 5

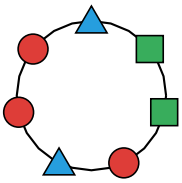
10. Tom has this bracelet with circles (red), squares (green) and triangles (blue) and wants to take away three **neighbouring** stones, one of each shape. In how many different places of the bracelet is this possible?

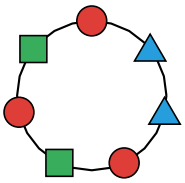


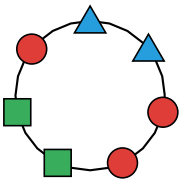
A. 2 ***B.*** 3 ***C.*** 4 ***D.*** 5 ***E.*** 6

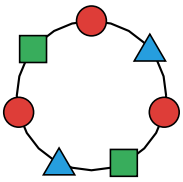
11. The picture shows a bracelet with circles (red), squares (green) and triangles (blue). Leo removes three neighbouring stones, one of each shape. Which bracelet can Leo **not** get?

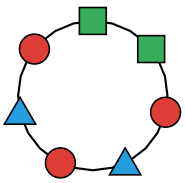


A. 

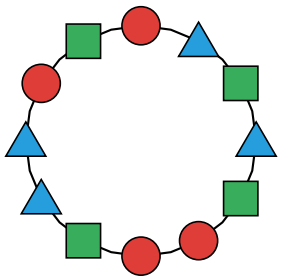
B. 

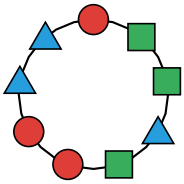
C. 

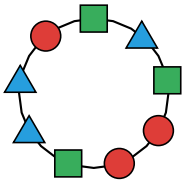
D. 

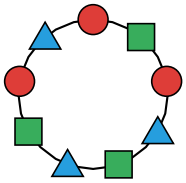
E. 

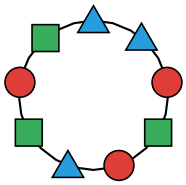
12. The picture shows a bracelet with circles (red), squares (green) and triangles (blue). Sara removes three neighbouring stones, one of each shape. Which bracelet can Sara **not** get?

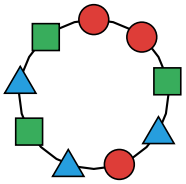


A. 

B. 

C. 

D. 

E. 

Answer Key

1. **D**	2. **A**	3. **B**	4. **E**	5. **A**	6. **C**
7. **C**	8. **D**	9. **C**	10. **C**	11. **A**	12. **A**

Solutions

1. D — Go around the bracelet and look at every group of 3 neighbours (also the ones across the top). Taking away the group circle, square, triangle leaves the bracelet in D, turned a bit. The other pictures need stones that were not next to each other, or other stones. [set-01/by_level/Grade_03-04_Ecolier/2025 Q21 · remove_three]

2. A — Go around the bracelet and look at every group of 3 neighbours (also the ones across the top). Taking away the group square, triangle, circle leaves the bracelet in A, turned a bit. The other pictures need stones that were not next to each other, or other stones. [set-01/by_level/Grade_03-04_Ecolier/2025 Q21 · remove_three]

3. B — Go around the bracelet and look at every group of 2 neighbours (also the ones across the top). Taking away the group circle, square leaves the bracelet in B, turned a bit. The other pictures need stones that were not next to each other, or other stones. [set-01/by_level/Grade_03-04_Ecolier/2025 Q21 · remove_two]

4. E — Go around the bracelet and look at every group of 3 neighbours (also the ones across the top). Taking away the group triangle, circle, triangle leaves the bracelet in E, turned a bit. The other pictures need stones that were not next to each other, or other stones. [set-01/by_level/Grade_03-04_Ecolier/2025 Q21 · remove_two]

5. A — Pick a special spot (for example a pair of equal stones) and read the stones in order from there, going around. Only A has all 9 stones in the same order; in the others two stones have swapped places. [set-01/by_level/Grade_03-04_Ecolier/2025 Q21 · same_bracelet]

6. C — Pick a special spot (for example a pair of equal stones) and read the stones in order from there, going around. Only C has all 10 stones in the same order; in the others two stones have swapped places. [set-01/by_level/Grade_03-04_Ecolier/2025 Q21 · same_bracelet]

7. C — Count each shape before and after: circles 2 → 1, squares 3 → 2, triangles 5 → 4. So Anna took away 1 circle, 1 square and 1 triangle. [set-01/by_level/Grade_03-04_Ecolier/2025 Q21 · which_removed]

8. D — Count each shape before and after: circles 5 → 3, squares 4 → 3, triangles 2 → 1. So Ben took away 2 circles, 1 square and 1 triangle. [set-01/by_level/Grade_03-04_Ecolier/2025 Q21 · which_removed]

9. C (3) — Go once around the bracelet and look at each group of 3 neighbours. Don't forget the group across the top: 1 of them. In all 3. [set-01/by_level/Grade_03-04_Ecolier/2025 Q21 · count_places]

10. C (4) — Go once around the bracelet and look at each group of 3 neighbours. Don't forget the group across the top: 1 of them. In all 4. [set-01/by_level/Grade_03-04_Ecolier/2025 Q21 · count_places]

11. A — Find all groups of 3 neighbours with one stone of each shape (also across the top) and take each one away. A has the right stones, but they were not next to each other. [set-01/by_level/Grade_03-04_Ecolier/2025 Q21 · cannot_make]

12. A — Find all groups of 3 neighbours with one stone of each shape (also across the top) and take each one away. A has the right stones, but they were not next to each other. [set-01/by_level/Grade_03-04_Ecolier/2025 Q21 · cannot_make]